AI-Assisted Legacy Modernization

Pilot report and proposal: a COBOL mainframe application translated to Rust with Claude Code

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In one slide

What we did

- Took a complete mainframe application: a credit-card system in **COBOL**, with online screens, nightly batch jobs and mainframe data files
- Had an AI coding agent, **Claude Code**, translate it into **Rust**, a modern, safe language
- Required **proof**: old and new programs run on the same data and must give the same output

What we got

- All **30 programs** translated, working and tested
- **40 of 40** batch outputs and **399 of 399** screens identical to the original
- **41 of 42** test cases passed, the last one's automated part too
- **15 defects** found in the original and documented
- About **6 hours** of AI working time, one engineer supervising

Recommendation: run a second pilot on one of our own legacy systems, with the same method.

Why legacy code is a business risk

The situation

- Core business rules live in decades-old COBOL, PL/I or RPG code
- The people who wrote it are retiring; few new engineers learn these languages
- Mainframe capacity and licences are a large, fixed cost
- Manual rewrites are slow and risky: rules are buried in the code, and small differences in rounding, dates or edge cases break the business

What a good solution must do

1. Understand the old code, *including its quirks*
2. Produce maintainable code in a modern language
3. **Prove** the new system behaves like the old one
4. Keep people in control of scope, review and release
5. Be fast and repeatable across many systems

The original application: what CardDemo is for

CardDemo is the core system of a **credit-card issuer**, the kind a bank runs on its mainframe: who the customers are, which cards and accounts they hold, what they spent and what they owe.

Who uses it

- **Bank staff**, through green-screen terminals: accounts, cards, transactions, bill payments, reports
- **Administrators**: which staff may sign on, and with which role
- **The nightly batch**: unattended programs that process the day's card activity

What it holds

- Customers, their accounts and credit cards
- Balances, credit and cash limits, billing-cycle totals
- Every card transaction: purchases, returns, payments, interest
- Transaction types, categories, interest-rate groups
- Staff user ids and roles

A day in the life of the system

During the day: online screens

- **Sign on**: menu by role
- **Accounts**: view, change limits and address
- **Cards**: list, view, block, edit
- **Transactions**: browse, view, add
- **Bill payment**: pay a balance in full
- **Reports**: request for any period
- **Users**: add, change, delete staff

At night: the batch cycle

1. **Posting**: check, post or reject the day’s transactions
2. **Interest**: per account and category
3. **Combine** daily and interest transactions
4. **Statements**: per customer, text and web
5. **Reports**: page, account and grand totals
6. **Export / import** for another branch

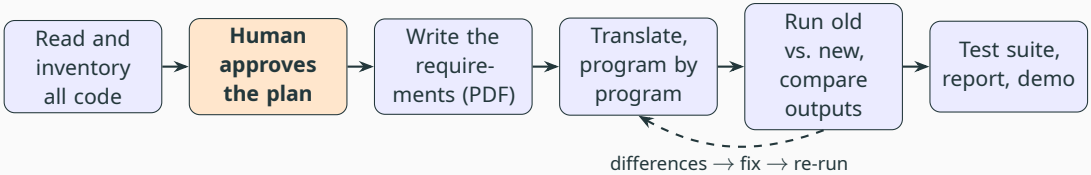
What was translated

	Original (COBOL, mainframe)	Translation (Rust)
Online screens	17 programs (IBM CICS), 17 screen layouts	17 modules + terminal screen engine
Nightly batch	13 programs + JCL job scripts	13 modules + job runner
Data	VSAM files, EBCDIC encoding, packed decimals	Embedded database, same record layouts and bytes
Size	≈ 27,000 lines	≈ 10,800 lines (incl. generated code)

The application is **AWS CardDemo**, a public sample built to represent real mainframe banking systems. The original code is unchanged in the repository (`cobol/`), next to the translation (`rust/`).

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How the AI worked: plan first, prove at the end



- Before writing code, the AI asked what to include. A person chose the scope: base application, terminal screens, embedded database, the original's behaviour kept exactly.
- Every phase ended in a saved, reviewable step (25 commits), with a written log of decisions.
- The AI also produced the documents people need: requirements, test suite, demo guide, this report.

Results at a glance

Measured results

Measure	Result
Programs translated	30 / 30
Batch outputs vs. COBOL	40 / 40 identical
Screens vs. COBOL	399 / 399 identical
Manual test cases	41 / 42 + 1 (step 2)
Defects found in the original	15
AI working time	≈ 5.9 h

Documents produced by the AI

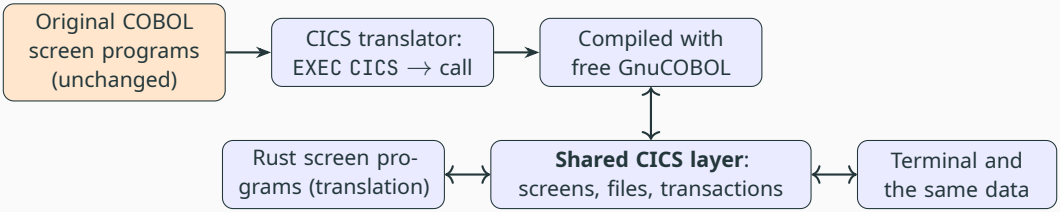
- Requirements specification: 28 pages, about 170 functional requirements
- Manual test suite with execution log
- Live demo guide for non-technical audiences
- Progress report with time and token figures
- `CLAUDE.md`: the rulebook for future AI work on this code

Proof: the old and new systems side by side

Nightly job	What is compared	Result
Daily posting	updated accounts, balances, transactions, rejected items	✓ identical
Interest calculation	accounts, generated interest transactions	✓ identical
Transaction report	519-line printed report	✓ identical
Customer statements	50 statements, text and web version (7,900 lines)	✓ identical
Data export / import	migration file, re-imported files, error file	✓ identical
File print utilities	5 programs, incl. packed-decimal outputs	✓ identical
Online screens	399 screens of 57 scripted terminal sessions: every field, colour, cursor	✓ identical

Original COBOL (free compiler) and Rust run on **the same data, clock and keystrokes**; one command each repeats the checks.

The original online screens, live



- Real CICS runs only on licensed mainframes: the original programs get a **stand-in for CICS**, the layer the Rust version already uses. Both runs share it, so the comparison checks the business logic
- All 17 original screen programs run **unchanged**; old and new screens can be shown **side by side**
- Automated comparison: **399 of 399 screens identical**; on the way it found 13 differences in the Rust port (all fixed) and 3 more defects of the original (documented)
- Differential fuzzing with random transactions and keystrokes found 7 more

Why not run it on an emulated mainframe?

The idea

Hercules, a free PC emulator of IBM mainframe hardware: why not run the original on it?

What a free setup can run

- MVS 3.8j, the operating system from 1981 that may be used freely
- No CICS: only the hobbyist look-alike **KICKS**
- Only a 1970s COBOL compiler (ANSI COBOL 74)

Why that does not work for CardDemo

- CardDemo is modern **Enterprise COBOL** (COBOL 85 and later: EVALUATE, END-IF, inline PERFORM, intrinsic functions); it does not compile with a COBOL 74 compiler
- Making it compile would mean **rewriting the original**, the opposite of what the demo must show
- Current z/OS with real CICS on a PC needs IBM's **licensed** test environment (ZD&T)

The CICS bridge runs the original code unchanged, at no licence cost.

What the AI found that people had missed

In the original system

Kept and documented as *known defects*:

- Last account's interest is never saved
- The report counts the last amount twice
- Card update wipes the security code
- Account update moves the group id
- Two programs do not compile
- … 15 in total

In its own translation

Found by running the original, and fixed:

- Batch comparison: 9 subtle differences
- AI **code review** (scoped by `CLAUDE.md`): 10 issues
- Online comparison: 13 differences
- All fixed; none was visible in normal tests

The value is not only in writing code fast, but in verifying it automatically.

CLAUDE.md: the project rulebook for the AI

What it is

A plain-text file in the repository, read by Claude Code at the start of **every** session:

- **Scope**: what may change
- **How to build, test, prove**
- **Review policy**
- **Lessons learned**

Excerpt

```
## Code review
Review only rust/.
- cobol/ is the original source and
  the specification: never report
  findings against it.
- A difference from the COBOL is a
  Rust bug. Known defects (KD-nn)
  are kept on purpose.
```

One reviewed, versioned rulebook keeps every session consistent.

Human oversight built into the workflow

Control points

- **Plan approval**: no code until a person accepted plan and scope
- **Checkpoints**: every step is a commit that can be reviewed or undone; nothing is published unasked
- **Budget limits**: the AI stopped at 85% of the usage quota and wrote a hand-over note; work resumed exactly there the next day
- **Memory and logs**: decisions and findings are written down

Claude Code Remote Control

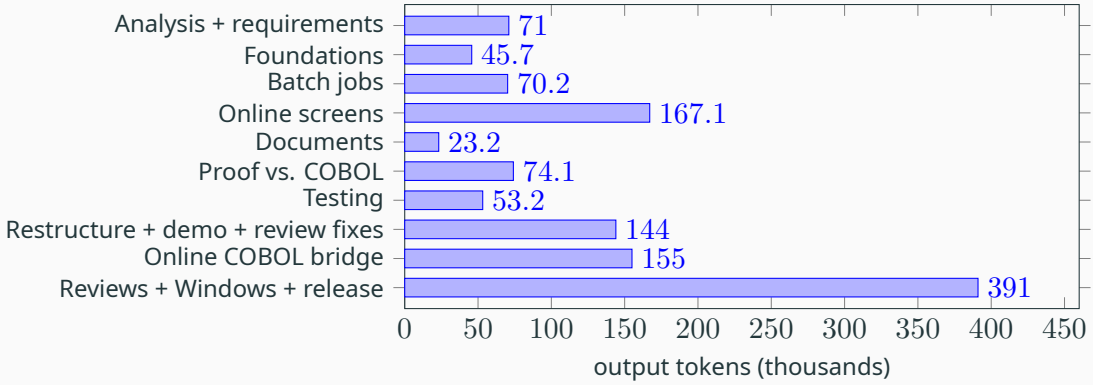
- The work runs on the company workstation; the code stays there
- From the **Claude mobile app**, the engineer follows the session, answers the AI's questions and approves steps while away from the desk
- Long runs no longer need someone sitting at the screen

Time and tokens

Session	Work	AI time	Output tokens
1	Analysis, requirements, translation of all 30 programs, test suite, report	1.1 h	367k
2	Proof against the original COBOL, test execution, fixes	0.5 h	137k
3	Restructure, demo tooling and guide, `CLAUDE.md`, code-review fixes, this presentation	0.9 h	144k
4	Original COBOL screens on a CICS stand-in, online comparison, fixes	0.6 h	155k
5	Eight code-review rounds, Windows build, terminal safety, downloadable demos	2.8 h	391k
Total		**≈ 5.9 h**	**≈ 1.2M**

- Measured from the Claude Code session logs (pauses over 10 minutes removed).
- Token cost depends on the subscription or API price list in use.
- The engineer’s own time went into decisions and reviews, not into typing code.

Where the effort went



Of the translation itself, the online screens were the largest part: 17 interactive programs with many validation rules each. The eight code-review rounds, the Windows build and the downloadable demos that followed took the most of all.

Risks and mitigations

Risk	Mitigation in this method
AI produces plausible but wrong code	Run old and new on the same data; outputs must be identical
Hidden rules in old code are lost	Requirements are written first, from the code, and reviewed
Original defects get “fixed” silently	Defects are kept and documented; fixing them is a business decision
Parts that cannot run off the mainframe (online CICS screens)	Screens drawn from the original layouts; 41-case test suite; next: comparison on a test mainframe
Source code confidentiality	Code stays on company machines; only what is sent to the model leaves them, under the company’s agreement with the vendor
Over-reliance on one tool	Output is ordinary Rust code plus documents; any team can maintain it

From pilot to programme

Next pilot (4–6 weeks)

1. Pick one of our legacy systems: bounded, with test data
2. Write its `CLAUDE.md`: scope, target stack, review rules
3. Set up the comparison harness: old vs. new on the same data
4. Translate, verify, document, as in this pilot
5. Review with the system owners; agree go-live criteria

What we need

- One engineer who knows the system, part time
- Access to the code and representative test data
- Claude Code licences for the team
- A test environment for the original system

Success criteria

Identical outputs on agreed test data, test suite passed, owners sign off.

Live demo

```
python3 demo/demo.py cobol      # the original COBOL programs
python3 demo/demo.py rust       # the Rust translation
python3 demo/demo.py compare    # → IDENTICAL, file by file
python3 demo/demo.py online-cobol    # original screens; online = Rust
python3 demo/demo.py online-compare  # → 399/399 screens identical
```

Step-by-step guide for presenters and non-technical audiences: `docs/DEMO.md`.

Try it yourself: the download

- GitHub, **Releases**: the Rust translation as one program with the sample data
- **Windows**: unpack the zip, **double-click** `carddemo.exe`
- **Linux**: unpack the tar.gz, run `./carddemo` (with GnuCOBOL: the original COBOL screens too)
- A menu: online screens, nightly batch, transaction report, output-file viewer
- No installation, no mainframe, no COBOL needed; instructions in `demo-quickstart.pdf`

Questions: Illia Burov

Appendix

Technical architecture

Component	Replaces	Lines
`cobol`	COBOL data rules: decimals, EBCDIC; records generated from copybooks	1,677
`vsam`	VSAM indexed files, on SQLite	556
`cics`	CICS screens (from the original BMS files), transfers, file control	833
`online`	the 17 online programs, one function per COBOL paragraph	4,283
`batch`	the 13 batch programs and JCL jobs	1,886
`carddemo`	the application: terminal screens, job runner	440

Verification: `rust/difftest/run.py` (GnuCOBOL comparison) and `cargo test` (21 suites, including the 41 manual cases as scripted terminal sessions and the multi-user tests).

Known defects of the original (selection)

ID	Where	Effect
KD-01	Interest calculation	the last account’s interest is never written back
KD-02/03	Transaction report	last amount counted twice; no final account total
KD-07	Card update screen	the card security code is wiped
KD-12	Export / import	the programs do not compile as shipped
KD-13	Account update screen	the group id is saved into the ZIP-code field
KD-14	Admin menu	after a not-installed option, the next input is ignored
KD-15	Account update screen	an account with a blank group never counts as unchanged

All 15 are in the requirements specification, with the exact behaviour the translation reproduces.